

Square Meal Feeds

In addition to producing traditional crops like alfalfa, corn, soybeans, and wheat, Idle Acres, a 1,000-acre, fifth-generation farm in central Minnesota, had developed a niche product: Square Meal Feeds horse biscuits. Harlan Anderson created the alfalfa biscuits as a healthier alternative to feeding traditional hay and grain. The biscuits (Figure 1) became a critical source of revenue for the Anderson family farm. But packing them into 40-lb bags was a time-consuming and labor-intensive process, and as labor was getting more expensive and harder to come by, Mark Anderson, the manager of Idle Acres, found the family had a difficult decision to make.



Figure 1. Square Meal Feeds biscuits for horses

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Anderson determined that a robot could both pack the bags and load them onto pallets, turning tasks that would normally take several workers a few backbreaking hours into an effortless 20-minute process. But the investment in this kind of automation would take years to pay off.

Meanwhile, the Andersons had a related problem. Their horse-feed biscuits were designed for and marketed to casual horse owners,

known as hobbyists, who often lack specialized expertise in horse nutrition. Square Meal Feeds made horse-feeding easy for these owners, giving them more time to enjoy their horses.

Anderson worried, though, that hobby horse ownership was declining in popularity, making the investment in automation even riskier.¹ A few of their customers raised racehorses, which required a slightly different product. If the Andersons could attract more racehorse owners, it could help offset the risk of declining hobby horse ownership, making the investment in automation more attractive. However, racehorse owners and their trainers were notoriously finicky about nutrition, often developing their own specialized feed formulas.

The Andersons went over the projections again and again, but found it hard to decide based on numbers alone. At the end of the day, it was a risk either way. If they didn't automate, and employees became too scarce or expensive, they might not be able to produce the biscuits. If they did invest in a robot, they might not sell enough biscuits to ever pay it off.

¹ 18- to 24-year-olds are less likely to own horses than older population segments in the U.S., and horse ownership is declining among this population.

Background

Swedish immigrants Sven and Christina Anderson founded Idle Acres in 1871 in Cokato, Minnesota, settling in the Upper Midwest for its quality soil and familiar climate. Four generations later, Harlan Anderson lived in the same farmhouse, farming the same land as his great-grandparents, with his sons Mark and Richard.

A Doctor of Veterinary Medicine as well as a farmer, Harlan Anderson treated a number of horses, often hobby horses kept by neighboring farmers. Over time, he noticed that many of the horses' medical problems were digestion related. Because horses have no gallbladder, which stores bile from the liver to aid in digestion, their stomachs are regularly exposed to bile. A horse's digestive system works best when the horse can graze, eating small amounts of grasses regularly over long periods of time.

Domesticated horses don't always have access to the grazing needed for healthy digestive systems, so many owners feed their horses hay, which is typically alfalfa or other grasses condensed into bales or rolled into cylinders. However, this has some disadvantages, including waste. Horses can trample hay into the ground, and the leaves, which contain the majority of nutrients, can blow away. In addition, if horses are given limited access to the hay at specific feeding times, they may end up with empty stomachs for part of the day – and overeat when they are given hay. To resolve these problems, Harlan Anderson packaged his alfalfa into a more condensed form for consumption, similar to dog or cat food, and Square Meal Feeds LLC was born.

Square Meal Feeds are about 1.25-square-inch biscuits—just dense enough that they do not blow away. The biscuits are soft, and digestion is easy. Horses can eat what they want when they want.

Square Meal Feeds not only solved a problem for horses—it also solved a problem for Idle Acres. Many farm products are commodities, meaning that they are undifferentiated in the market. Prices for these products are determined at the national or even global level, based on supply from all competing farmers and demand from food producers and customers, like General Mills. No one farmer has pricing power over the grains they sell, which makes it hard to achieve profits.

The Andersons' lack of pricing power was compounded by conditions facing Midwestern farmers. While fewer people are farming, some farms have gotten significantly larger. Larger farms are better able to scale, making it easier to achieve profits by lowering production costs.

Minnesota alone lost 5% of its farms between 2017 and 2022. While the Andersons' 1,000-acre farm was larger than the average Minnesota farm of 379 acres (USDA 2023), it wasn't nearly large enough to compete with much larger farms. In 2022, 27,000 U.S. farms were 5,000 acres or larger, representing 42% of all farmland. Meanwhile, nearly 1.9 million farms accounted for the remaining 58% of farmland (USDA 2024).

Having a differentiated product like Square Meal Feeds provided a much-needed revenue boost for Idle Acres and the Andersons. Square Meal Feeds produced three core products distributed nationally, with their primary distribution network in the Upper Midwest and the majority of sales coming from east of the Rocky Mountains:

- Next Generation: Formulated for foals up to two years of age
- Happy Horse: A maintenance product for adult horses
- Senior Supper: For older horses with poor teeth

A fourth product, Winner's Circle, was developed for racehorses, but the Andersons weren't marketing it widely, in part because it would require hiring a salesperson just to deal one-on-one with owners and trainers. Harlan Anderson was also concerned that racehorse owners would demand a product with more protein content and oats than he believed was healthy.

The Decision

Mark Anderson, along with his father, Harlan, and brother, Richard, faced a tough decision. Investing in a robot to automate the process of filling bags and packing pallets would instantly eliminate the stress of relying upon a steady supply of labor. Anderson was already paying farmhands more than \$20/hour to compete with wages in other industries, and he had found it harder to find workers willing to do physically demanding manual labor.

The upfront cost for a robot would be \$225,000, financed at current rates (7%) over a five-year loan. But according to his calculations, it would take eight years for the investment to pay off,² and that's only if Idle Acres maintained or grew sales. Horses are notoriously expensive, requiring land on which to keep them and hundreds or even thousands of dollars in feed, nutritional supplementation, and veterinary care per year. Would the next generation embrace this hobby? Would the investment be worth it?

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² From an investment perspective, the robot would pay off once the cost of owning and operating the robot became less than the cost of hiring labor to do the job.

References

- LifeData. 2018. “Understanding the Horse’s Digestive System.” Life Data Blog, July 12.
<https://lifedatalabs.com/blog/2018/07/12/understanding-the-horses-digestive-system/>.
- Mallory, Beth. 2024. “Horse Industry Statistics in 2024 (Latest U.S. Data).” Horses Only, January 16. <https://horsesonly.com/horse-industry/>.
- USDA. 2023. “Minnesota Ag News – Farms and Land in Farms.” Press release, posted by National Agricultural Statistics Service, February 17.
https://www.nass.usda.gov/Statistics_by_State/Minnesota/Publications/Other_Press_Releases/2023/MN-Farms-02-23.pdf.
- USDA. 2024. “Farms and Farmland.” National Agricultural Statistics Service, March.
https://www.nass.usda.gov/Publications/Highlights/2024/Census22_HL_FarmsFarmland.pdf.